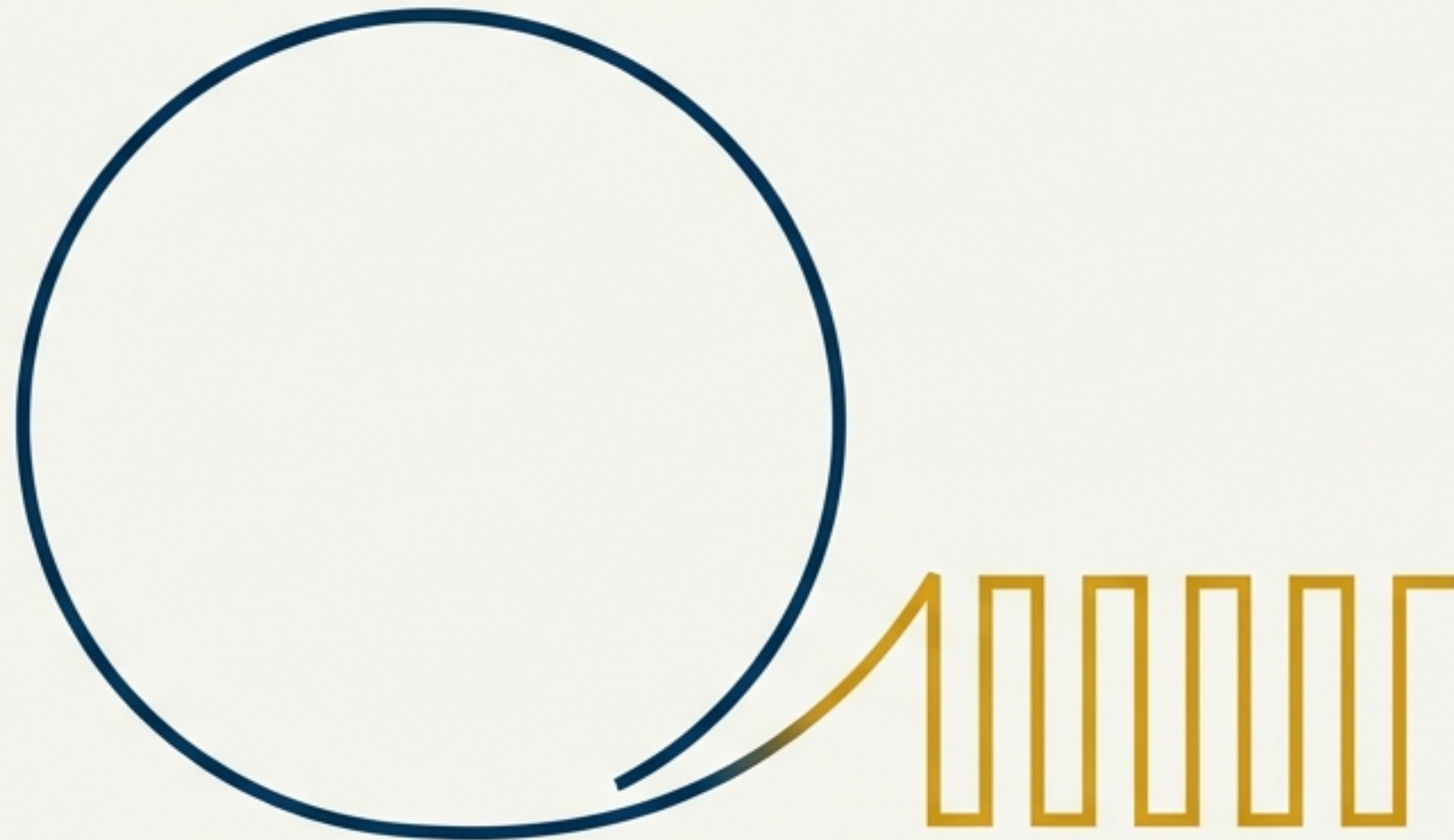


Delay-Induced Mode Discreteness

A Classical Route to Pattern Formation in Nonlinear Ring Systems

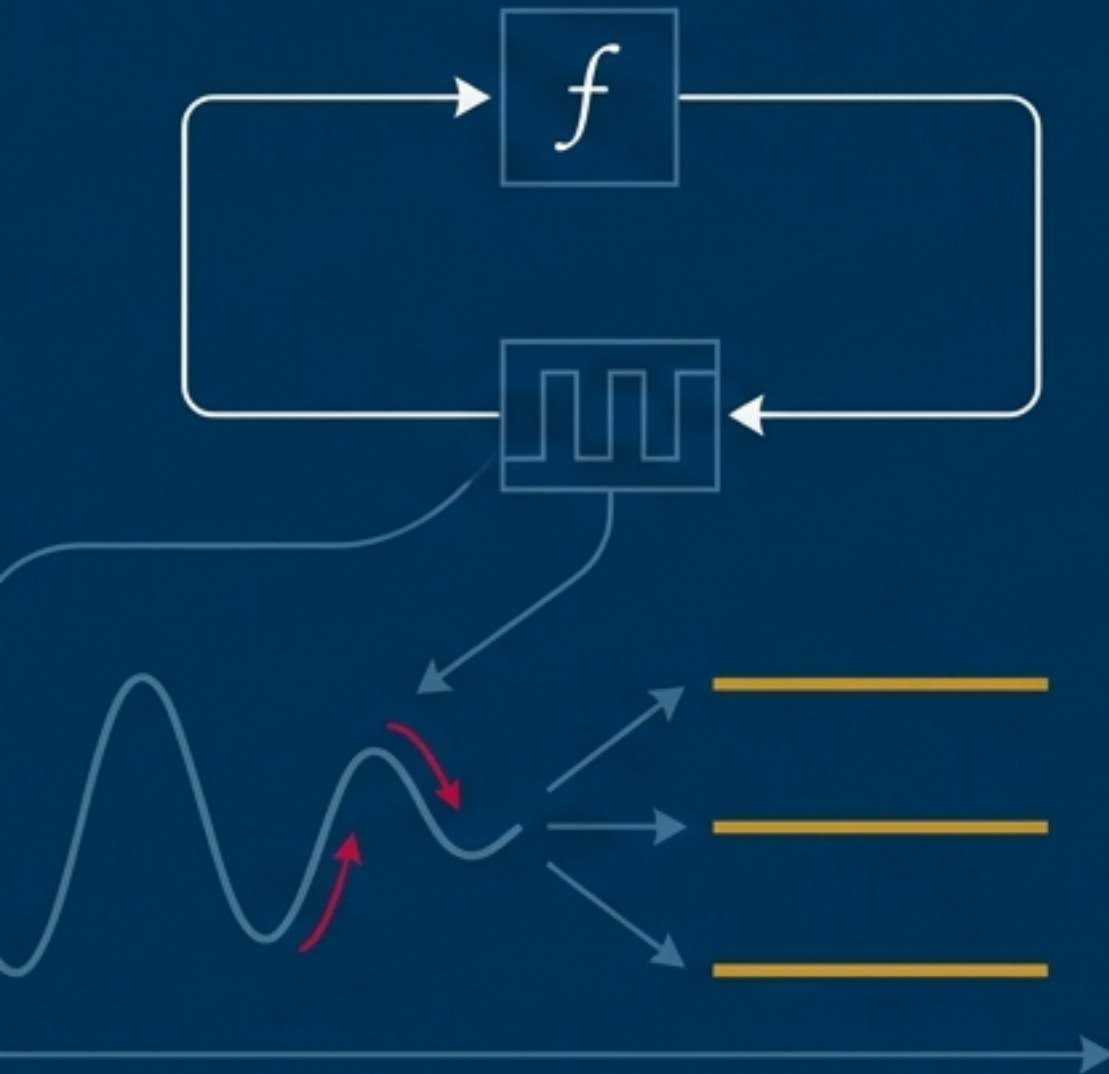


Based on the research of Omar Iskandarani | Independent Researcher, Groningen, The Netherlands | January 23, 2026

"Nonlinear delayed-feedback systems can display spatio-temporal organization even when governed by a single scalar delay differential equation."

Executive Summary

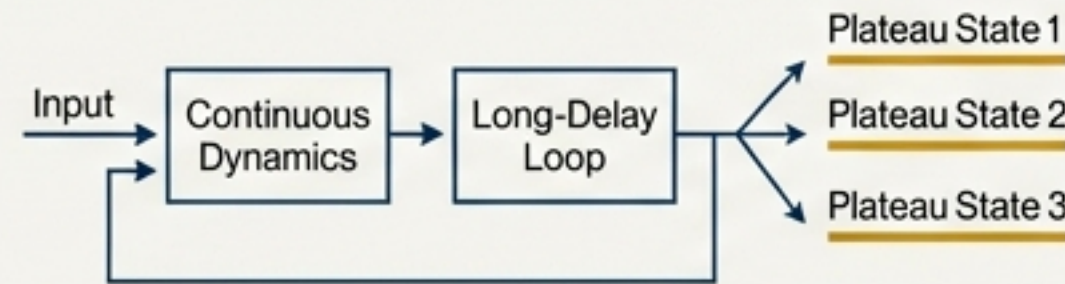
The core argument in brief.



Discrete states can emerge from classical continuous dynamics.

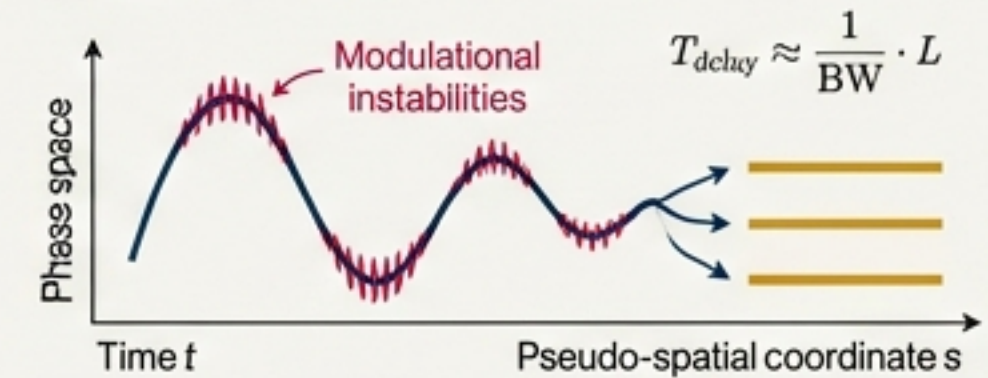
1 The Core Thesis

Long-delay feedback loops can spontaneously self-organize into discrete "plateau" states purely through classical dynamics, without requiring microscopic quantization assumptions.



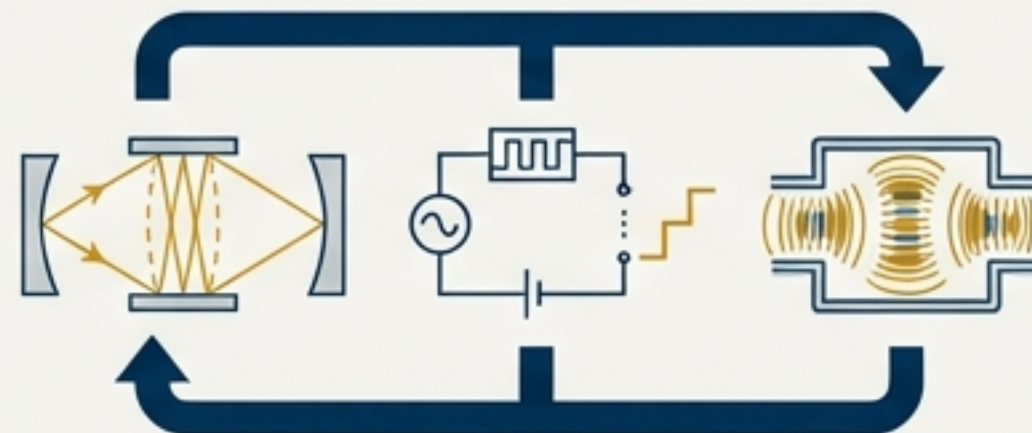
2 The Mechanism

The finite loop transit time defines an effective pseudo-spatial coordinate. Modulational instabilities on this coordinate select robust plateau-like states.



3 The Impact

Offers a unified framework for understanding mode selection in optical cavities, electronic delay loops, and microwave resonators.



4 Key Concept

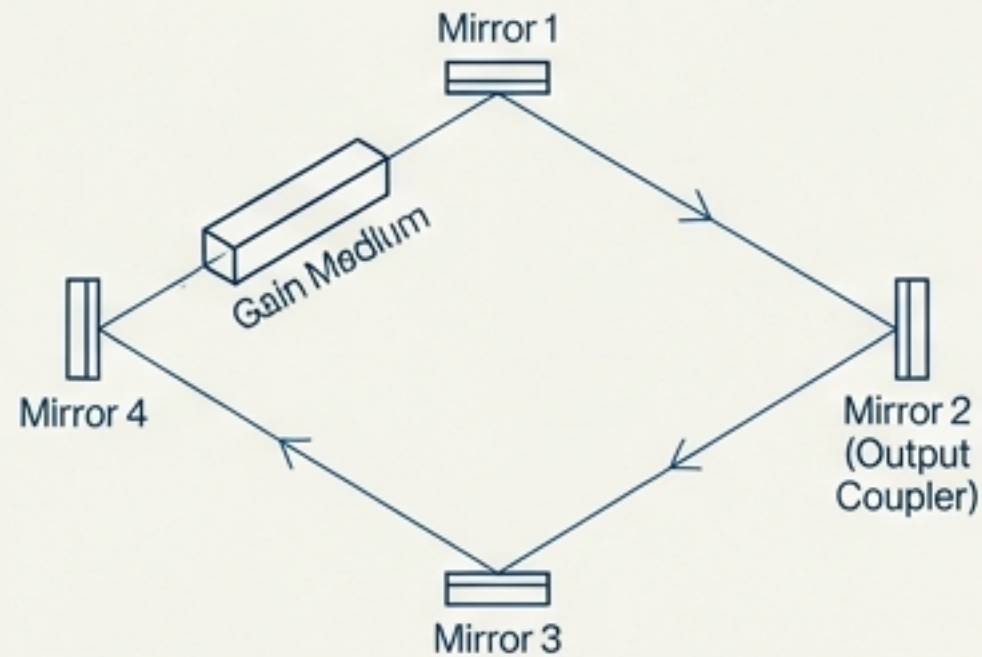
Distinguishes between *selection* (driven by delay dynamics) and *persistence* (analogous to Hamiltonian transport barriers).



The Context: Ubiquity of Nonlinear Ring Systems

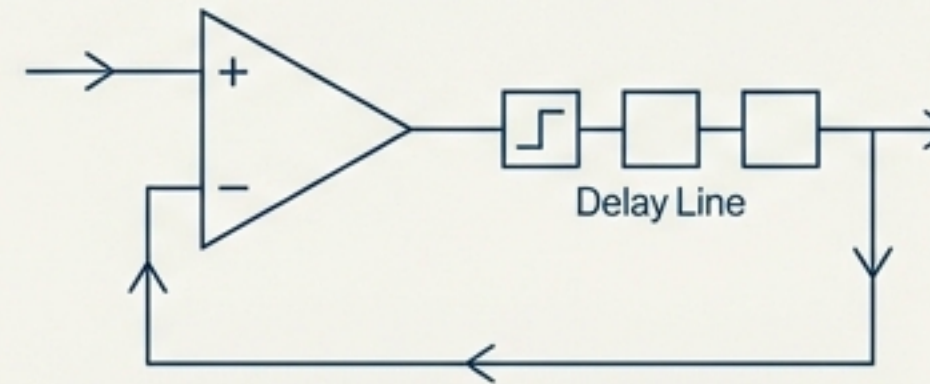
Diverse physical systems share a fundamental architecture: a signal circulating in a loop with feedback.

Optical Cavities



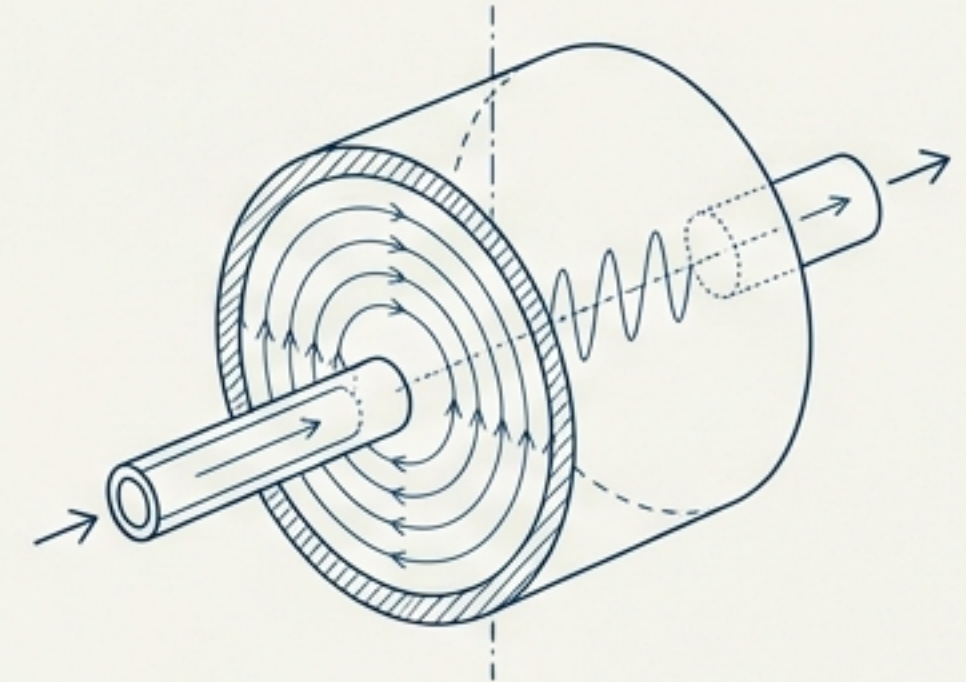
Lasers with delayed feedback loops.

Electronic Delay Loops



Circuits with delayed signal processing.

Microwave Resonators



Electromagnetic cavities and waveguides.

The Common Thread: Regardless of the medium (light, electrons, microwaves), the physics are governed by the interplay of nonlinearity and propagation delay.

The Central Tension: Continuity vs. Discreteness

The Inputs (Math)



Governed by continuous differential equations.



The Outputs (Observation)

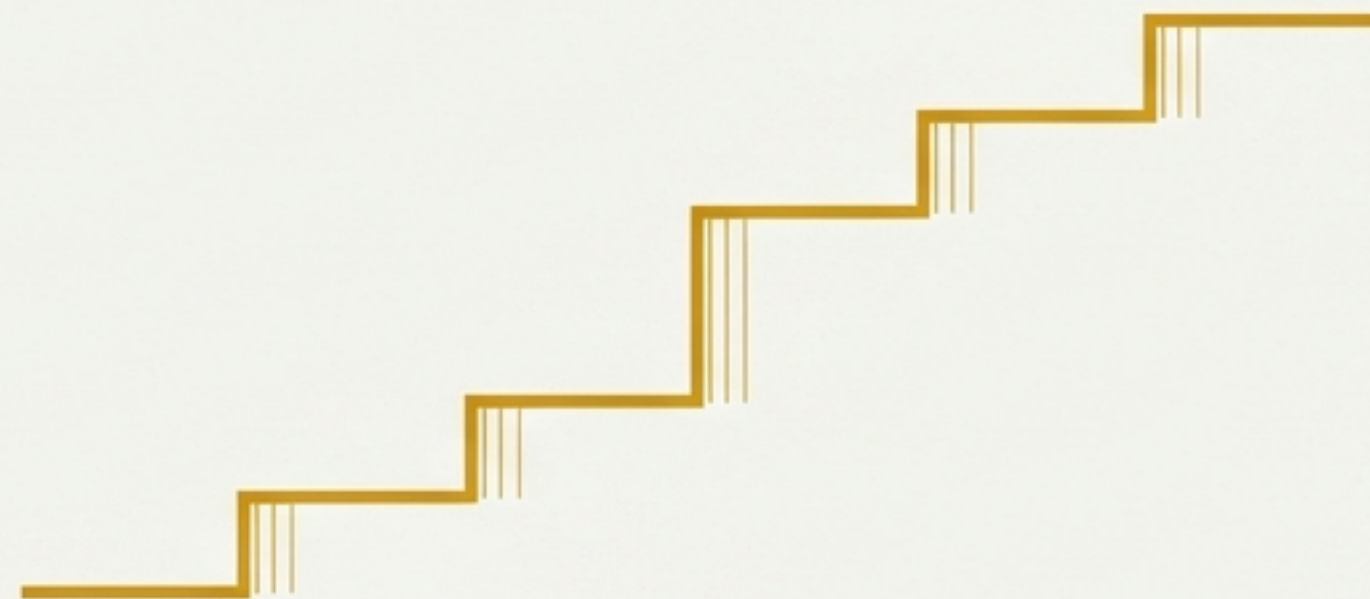


Exhibit distinct, discrete oscillation modes.

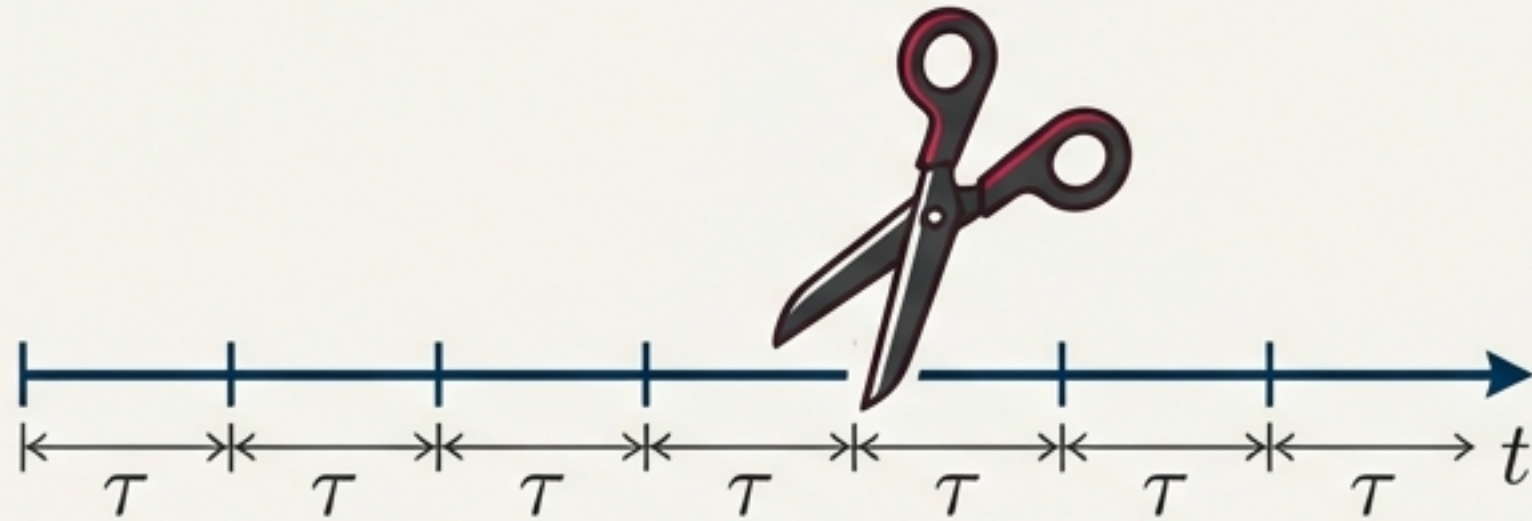
How does **discreteness** emerge from **continuity** without assuming microscopic quantization?

The Hypothesis: The “Long-Delay Limit” provides a purely classical explanation through self-organization.

The Hypothesis: In the Long-Delay Limit, Time Acts as Space

The Spatio-Temporal Analogy

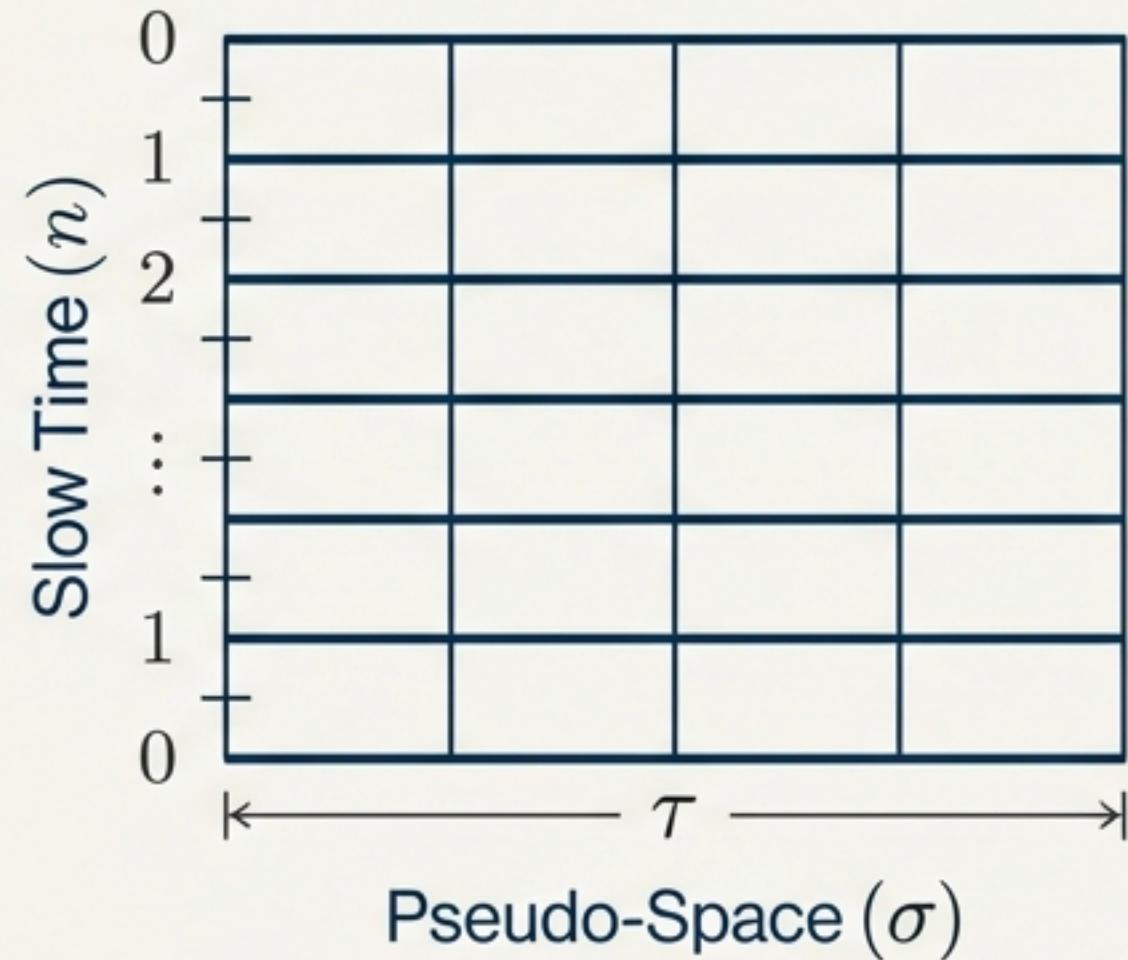
Continuous Time



Stacking



Pseudo-Space Map



When circulation delay (τ) $\gg$ relaxation time (ϵ),
the loop transit acts as a **physical spatial dimension**.

A Minimal Model for Nonlinear Delay Dynamics

The diagram shows the differential equation $\epsilon \dot{x}(t) = -x(t) + \mu x(t - \tau) - x(t - \tau)^3$ with four annotations connected by lines to specific terms:

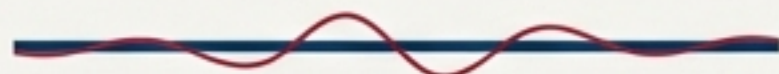
- Inertia** (Fast relaxation time, $\epsilon \ll 1$) points to $\epsilon \dot{x}(t)$.
- Damping (Loss)** points to $-x(t)$.
- Linear Feedback** (Gain, controlled by μ) points to $\mu x(t - \tau)$.
- Nonlinear Saturation** (Cubic term limiting growth) points to $-x(t - \tau)^3$.

$$\epsilon \dot{x}(t) = -x(t) + \mu x(t - \tau) - x(t - \tau)^3$$

This **single scalar delay** differential equation captures the essential physics of ring systems.

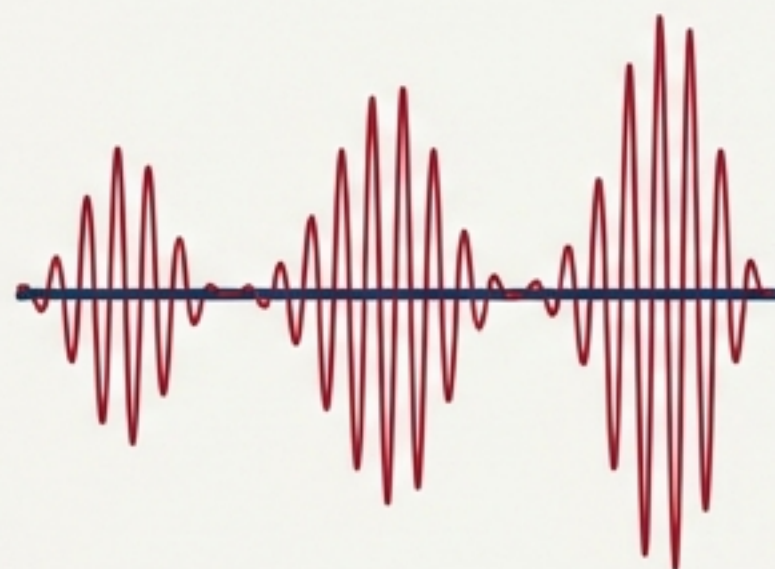
Modulational Instability Drives Symmetry Breaking

Stage 1: Instability



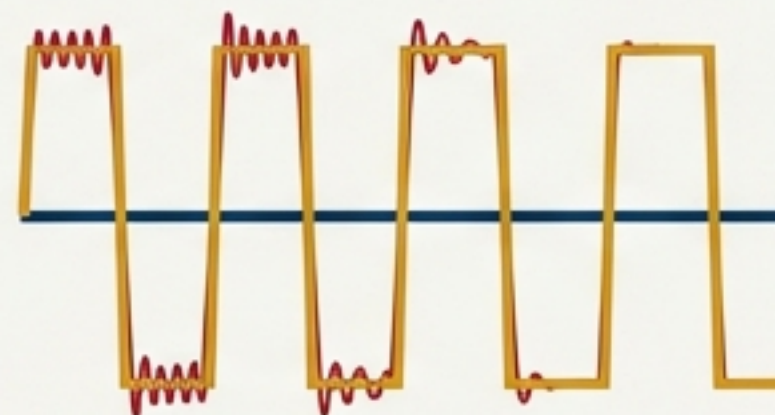
Uniform oscillations become unstable.

Stage 2: Organization



Small perturbations grow and interact via feedback.

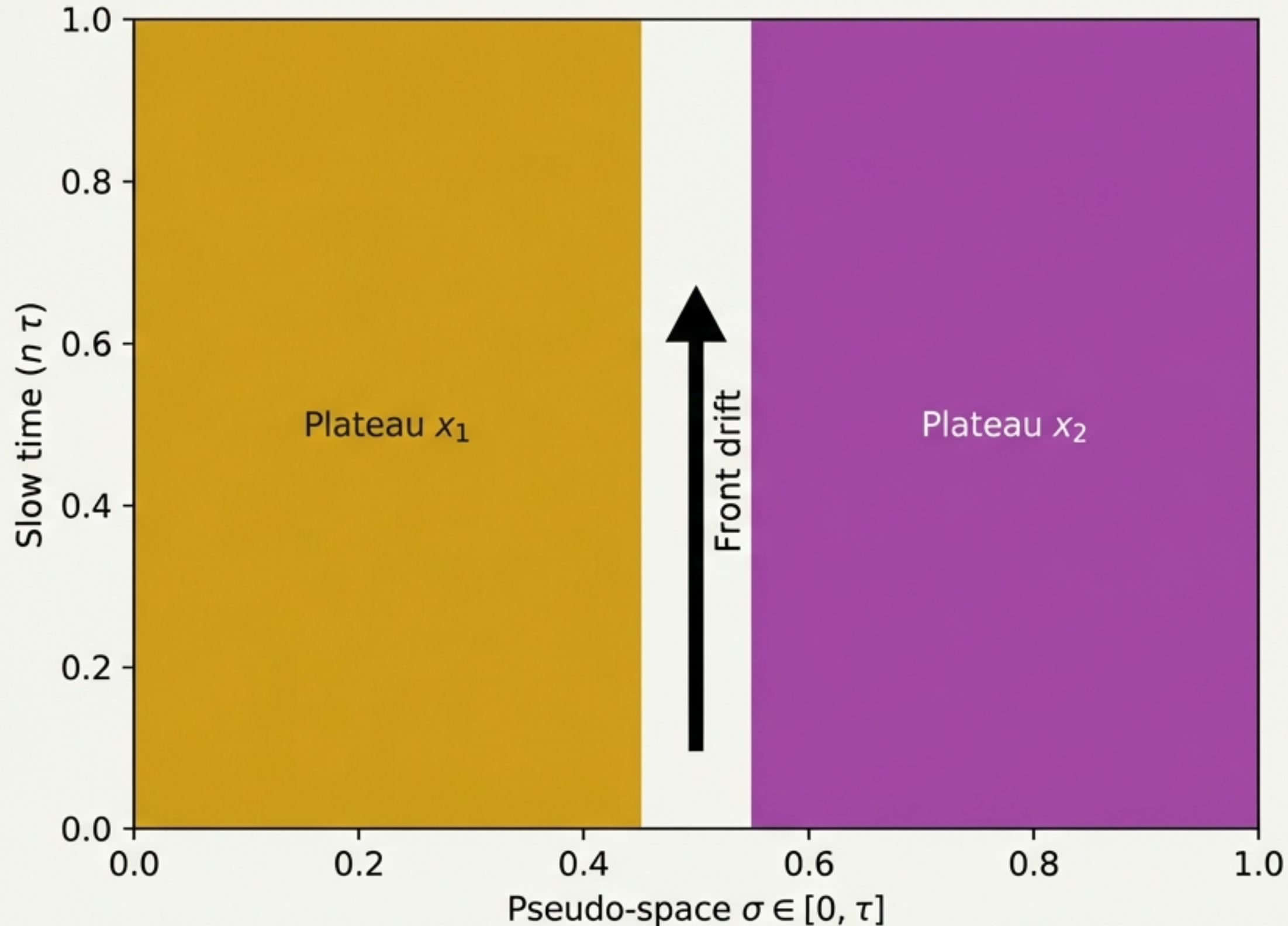
Stage 3: Selection



System breaks symmetry, selecting "plateau" states.

Note: In the long-delay limit, eigenvalues organize into branches, creating a continuous pseudo-spatial wavenumber (k). The dynamics select the pattern.

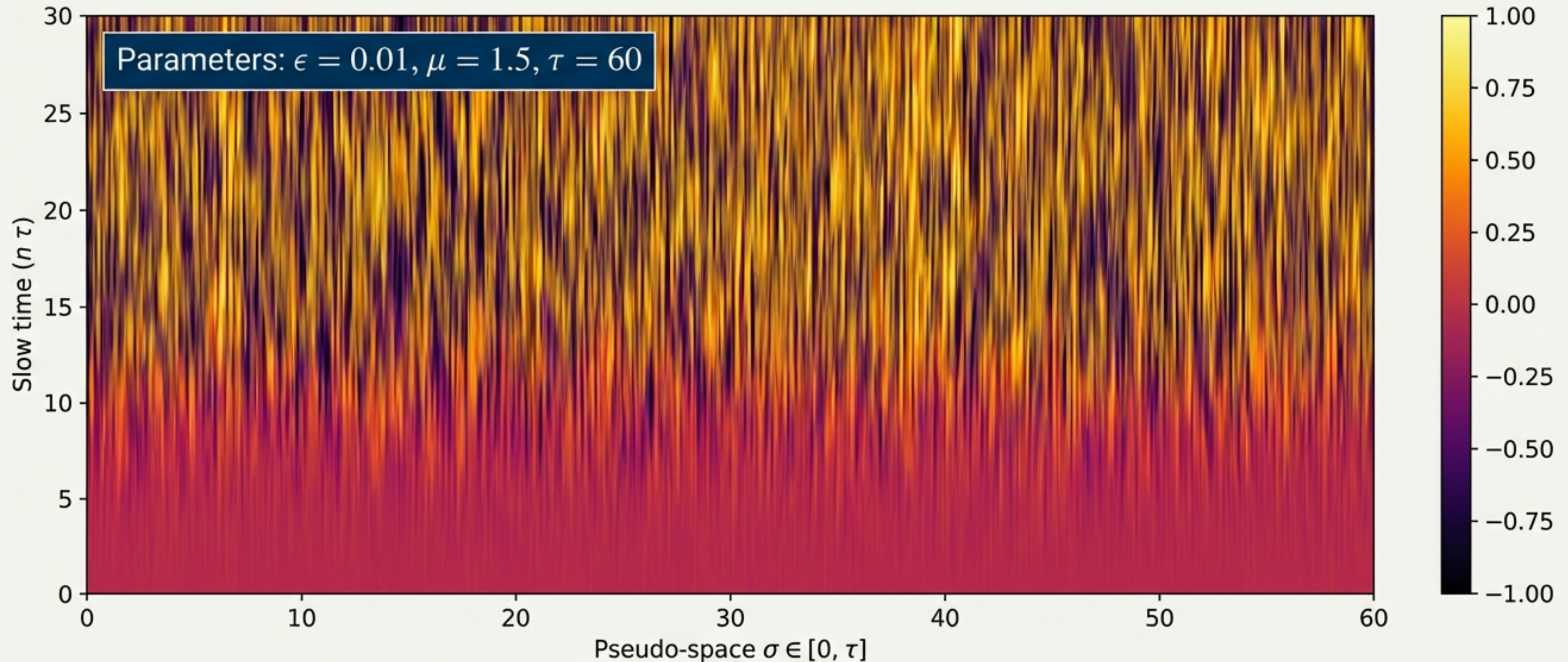
Visualizing the Dynamics: The Schematic View



Key Features

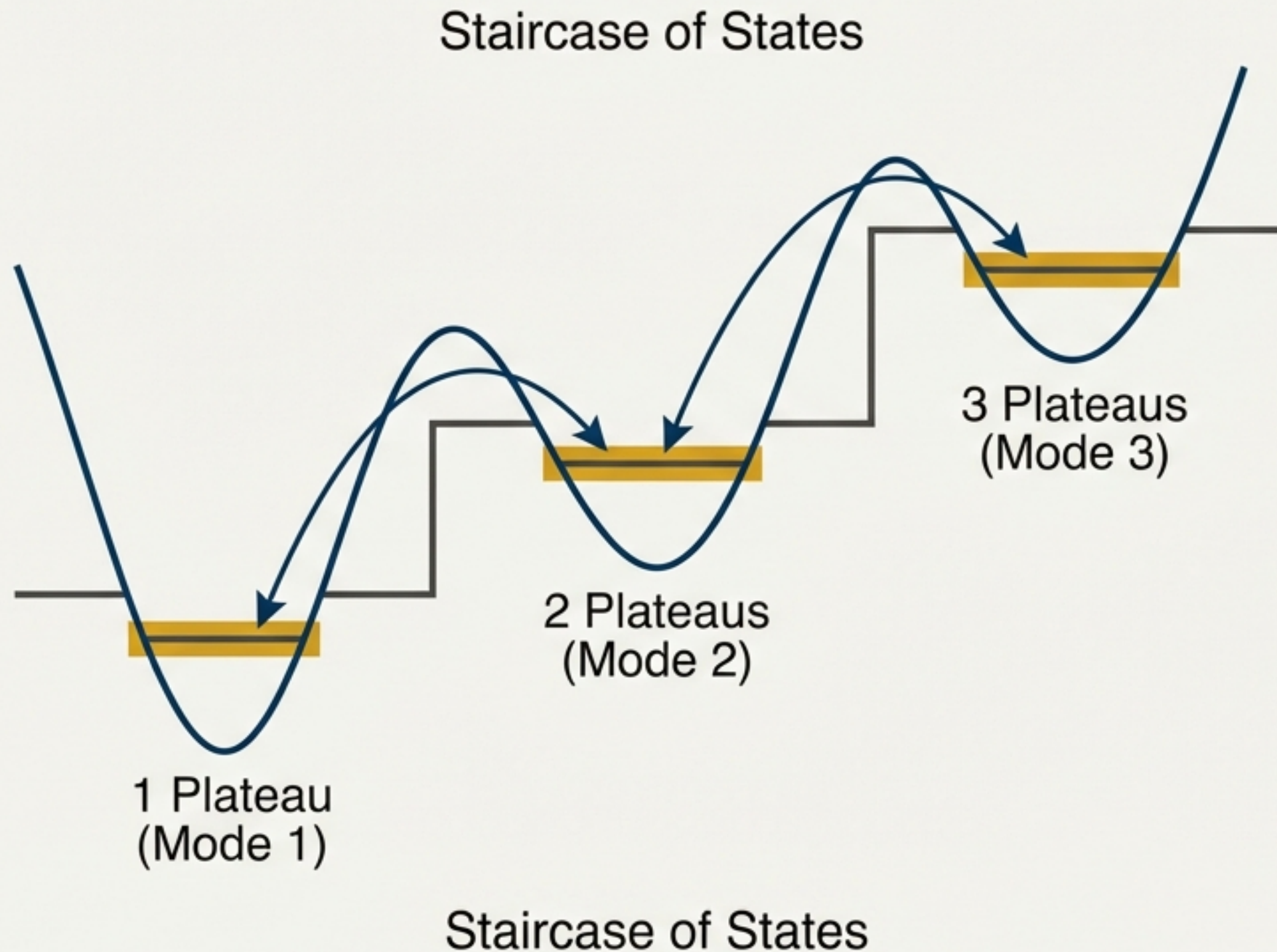
- **Plateaus:** Two flat levels (x_1 and x_2) representing bistable states.
- **Fronts:** Sharp transition layers (domain walls).
- **Drift:** The front moves slowly over slow time ($n\tau$), indicating mode selection.

Numerical Evidence Confirms Spontaneous Organization



Direct output of Eq. (1). The system spontaneously generates alternating plateau regions (vertical stripes) and sharp domain walls from a random initial state.

Multistability Implies Effective Discrete Modes



Multistability: For the same parameters $(\mu, \tau/\epsilon)$, the system can settle into states with different numbers of plateaus.

The Connection: These distinct plateau counts act effectively as 'Circulation Mode Numbers'.

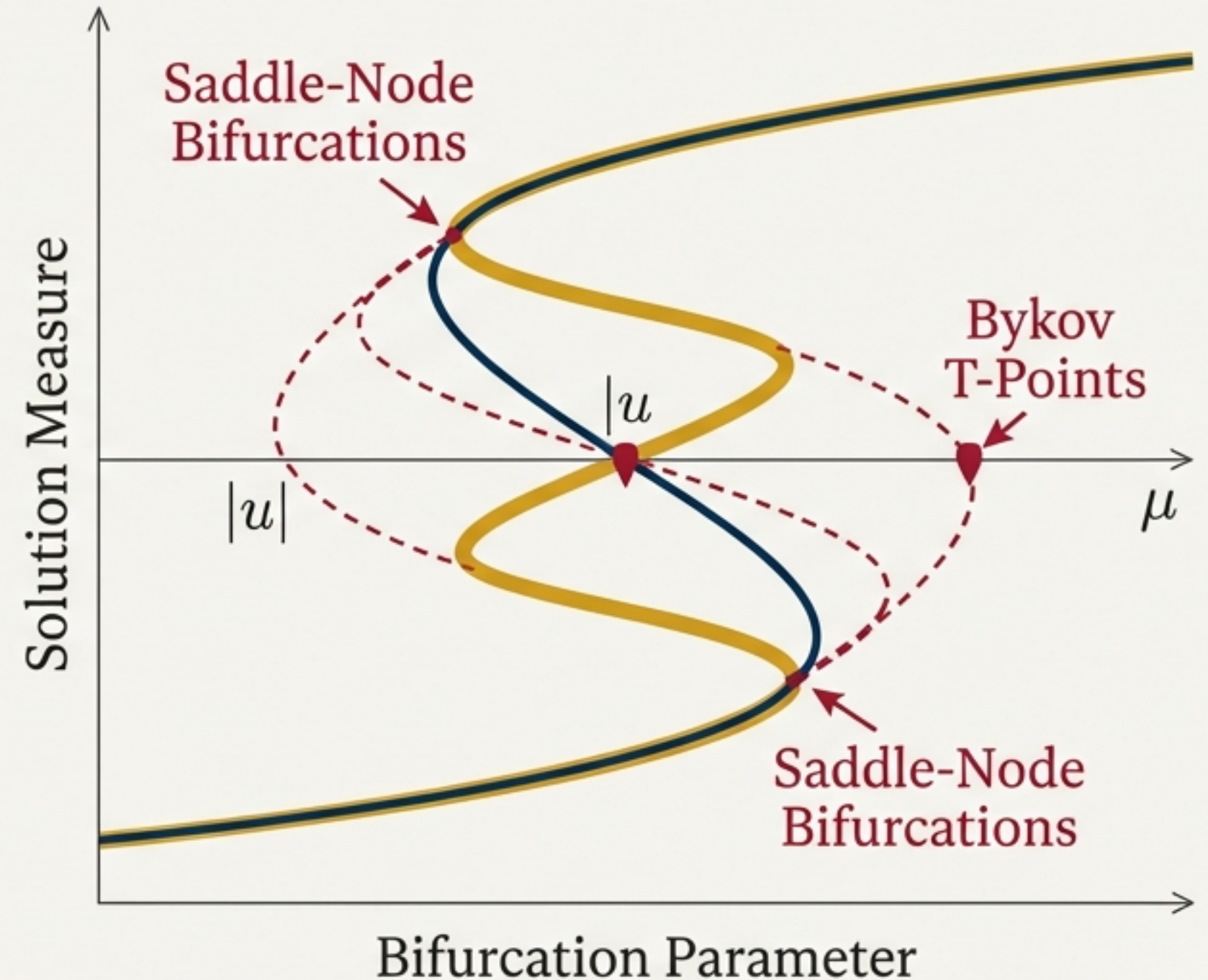
Significance: This mimics quantum numbers but arises purely from the classical delay length and feedback nonlinearity.

Global Topology & Bifurcations

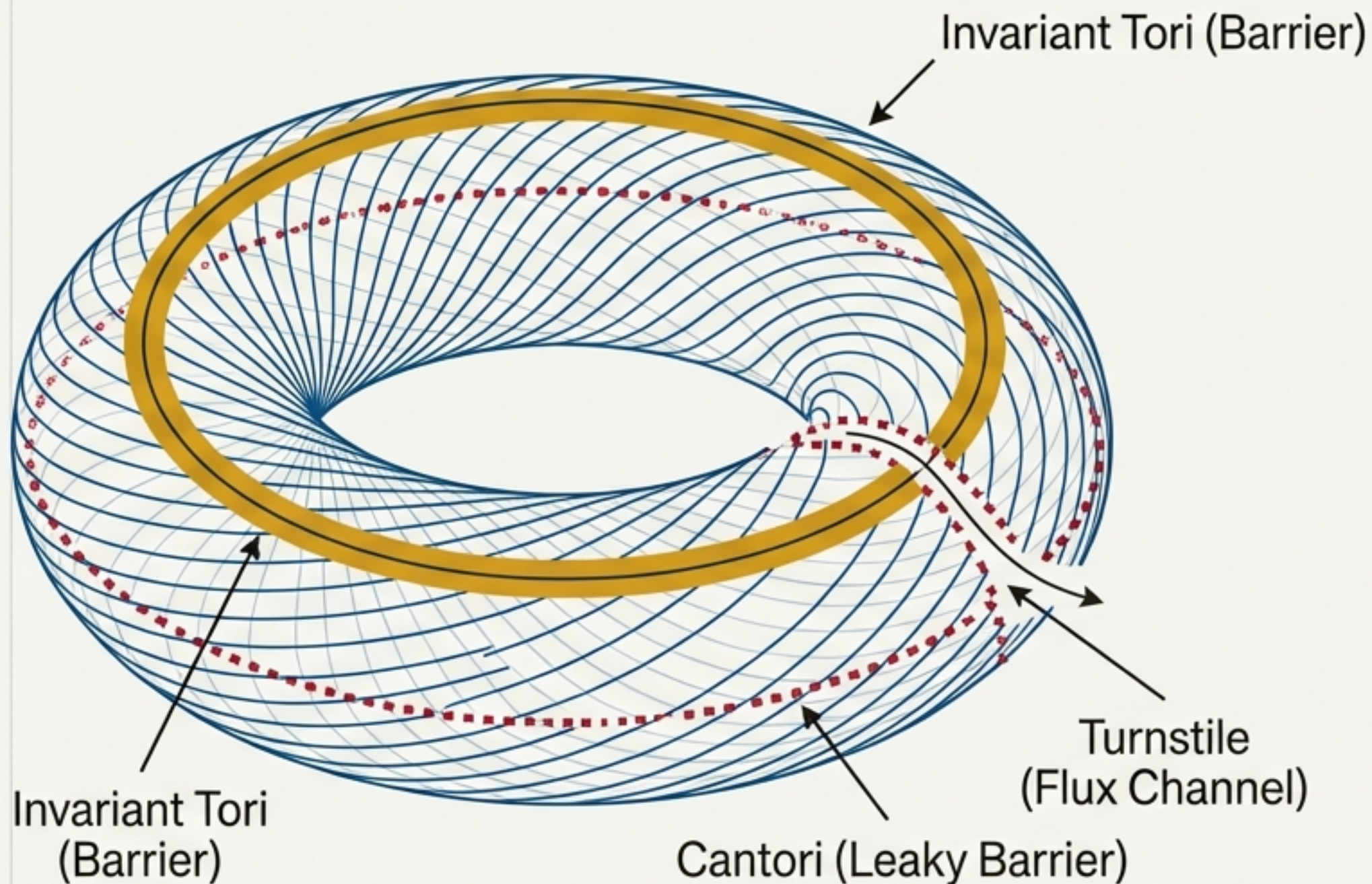
The Rules of the Game: The creation and destruction of modes are **not random**. They are governed by topological organizing centers.

Snaking Structures: Intertwined solution branches allow multiple modes to coexist.

Bykov T-Points: Global connections that organize the hierarchy of square-wave families.



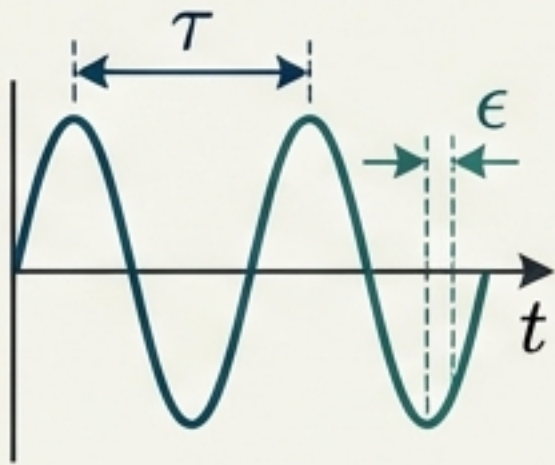
Why Do These Modes Persist? A Hamiltonian Analogy



- **The Analogy:** Circulating fields behave like trajectories in a toroidal phase space (similar to magnetic fields in a tokamak).
- **Topological Protection:** Modes are robust because they are trapped by “Cantori”—barriers that are hard to cross.
- **Decay:** Transitions only happen through rare, narrow channels called “Turnstiles”.

Scope & Limitations

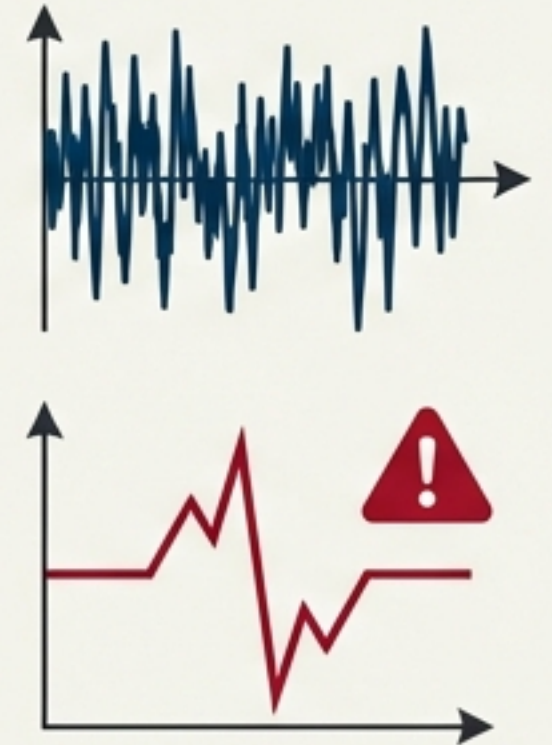
Valid Regime



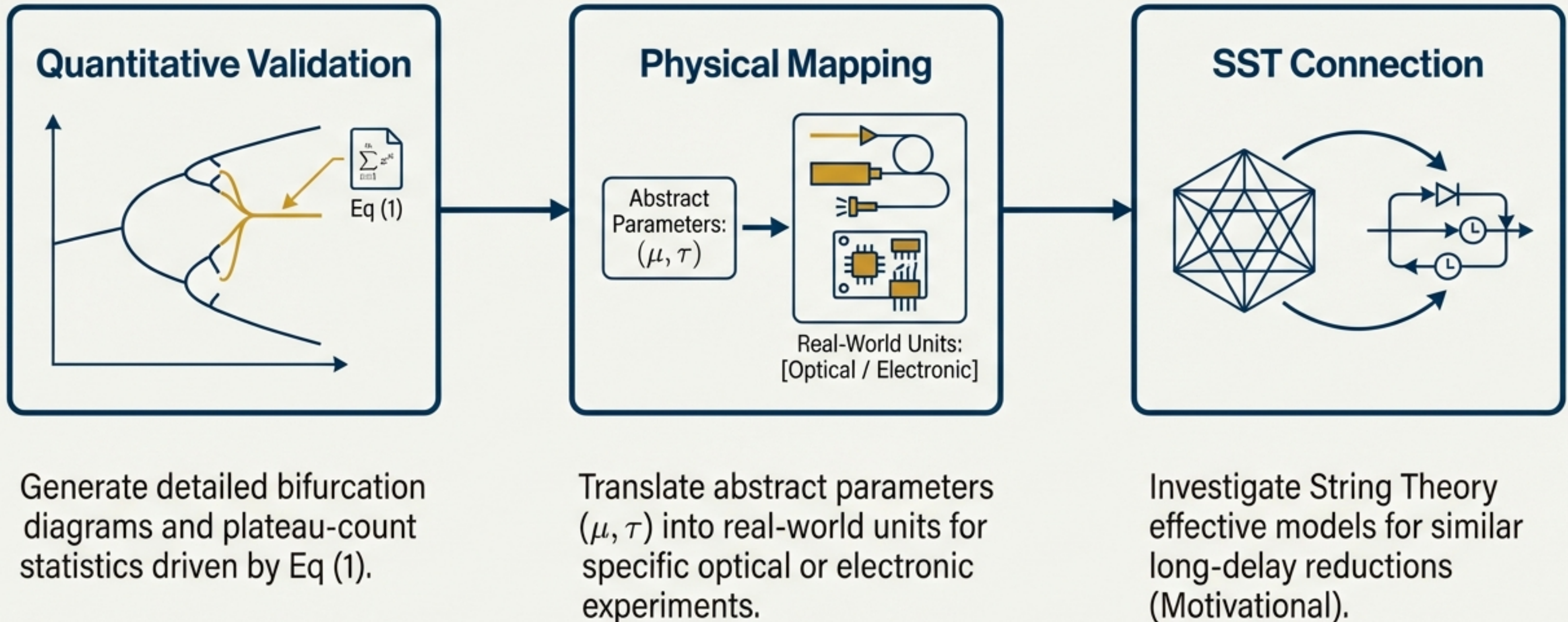
- **Long Delay:** $\tau/\epsilon \gg 1$ (Delay is dominant).
 - **Smoothness:** Requires smooth local nonlinearities.
- **Determinism:** Works best with weak noise.

Breakdown Points

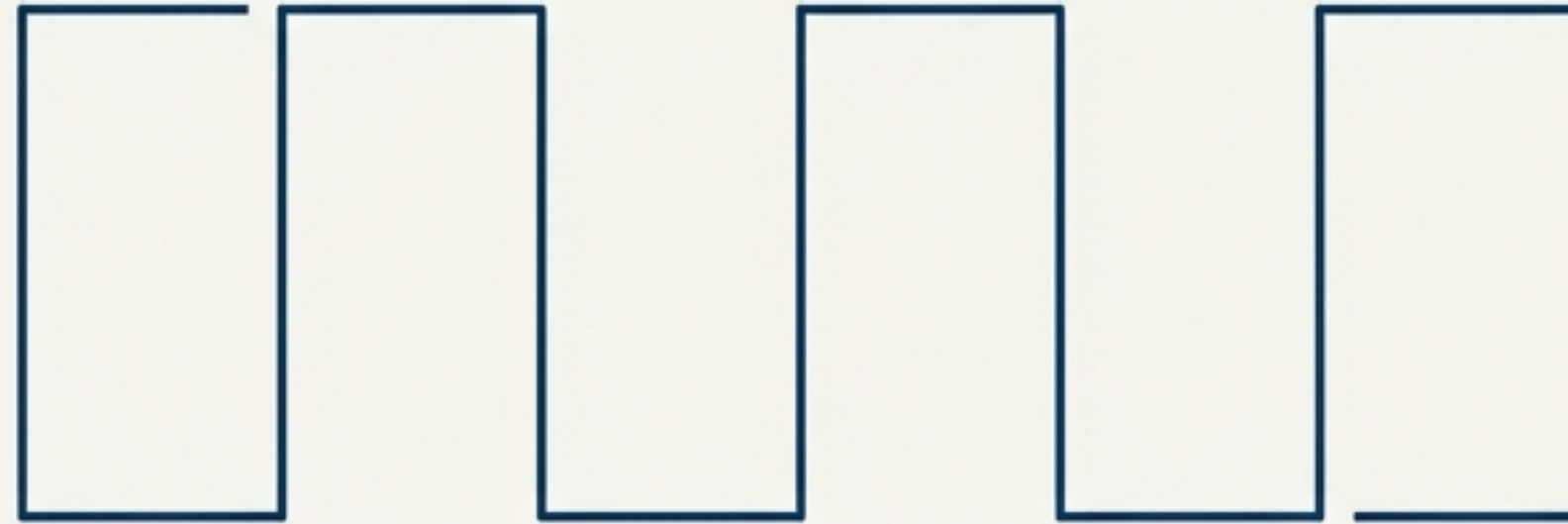
- **Strong Stochasticity:** High noise breaks the spatial analogy.
 - **Non-Smooth Systems:** Sharp discontinuities violate the model.
- **Note:** The Hamiltonian connection is a dynamical *analogy*, not a strict derivation.



Outlook and Future Research



Conclusion: A Generic Mechanism for Discreteness



1. We transformed a time-delay problem into a spatial pattern problem using pseudo-space (σ).
2. Delay-induced pattern formation is a robust, classical mechanism for generating discrete modes.
3. This framework explains 'quantized' behaviors in macroscopic systems without quantum postulates.

Explore this framework for interpreting data in photonic and electronic experiments.